

1 Theory of Computation Automata Part Questions

- 1) Explain different set theory operations with examples.
- 2) Tell different types of propositional logic gates and their truth tables.
- 3) What are different types of functions. Explain them.
- 4) Describe ~~the~~ KLEENE'S THEOREM and explain with examples.
- 5) What is MINIMAL FINITE AUTOMATA. DIFFERENT EXAMPLES OF PUMPING LEMMA TO PROVE THAT THE LANGUAGES ARE NOT REGULAR.
- 6) DESCRIBE CONTEXT FREE LANGUAGE, CONTEXT FREE GRAMMAR, PRODUCTION, UNIT PRODUCTION, NULL PRODUCTION WITH EXAMPLES.
- 7) DESCRIBE SYNTAX OF PROGRAMMING LANGUAGES. WHAT IS DERIVATION TREE OR PARSE TREE. EXPLAIN WITH EXAMPLES.
- 8) EXPLAIN CONTEXT FREE GRAMMAR OF A REGULAR LANGUAGE. EXPLAIN CHOMSKY NORMAL FORM. EXPLAIN WITH EXAMPLES.
- 9) DESCRIBE A PUSHDOWN AUTOMATA ACCEPTING THE LANGUAGE OF PALINDROMES.
- 10) DESCRIBE A TOP DOWN PDA AND DETERMINISTIC PDA (PUSHDOWN AUTOMATA) WHERE MORE ~~than~~ ^{less than} 1 symbol.

- 1) Tell context free grammar corresponding to a given PDA
- 2) What is different types of parsing. Explain them
- 3) PUMPING LEMMA EXPLAIN THE EXAMPLES ARE NOT CONTEXT FREE. ($a^i b^j c^k \mid i, j, k \geq 1$) is not context free
- 4) Using Ogden's lemma $\{a^i b^j c^k \mid i \neq j\}$ is not context free.
- 5) A context free language whose complement is not context free language.
- 6) Describe Turing Machine and Acceptance of Turing Machine. Universal Turing Machine
- 7) What is Recursively Enumerable languages and recursive language.
- 8) What are unsolvable problems. A NON RECURSIVE LANGUAGE AND AN UNSOLVABLE PROBLEM.
- 9) Explain the halting problem. Prove halting problem is unsolvable.
- 10) Explain Rice theorem and Post correspondence problems.
- 11) Explain primitive recursive function.
- 12) Explain Time complexity and Space complexity

3) What is context free grammar
 Explain context free grammar
 2) Prove that $\{a^n b^n \mid n \geq 1\}$ is context free

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tree grammar corresponding to
each type of parsing. In brief term
MA EXPLAN THE EXAMPLES ARE
(a) $\frac{1}{2} \frac{1}{2}$ / 177 not ordered.

Q. 3. What is CNF Satisfiability problem.
Explain Cook's Theorem.

24) PROVE That vertex cover problem is NP COMPLETE

25) Prove that Kolomibility problem is NP complete